

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<tool_call> reasoning process here

At birth , the brain undergoes rapid development , with different regions developing at different rates . The brain stem , thalamus , hypothalamus , and other subcortical structures are relatively well-developed at birth because they are essential for basic life functions such as breathing , heart rate , and sleep regulation . These structures are involved in survival and are thus developed early .

In contrast , the cortex , particularly the prefrontal cortex (which includes the frontal lobe) , is the most developed region in the mature brain but is the *least developed at birth* . The frontal lobe , which is critical for executive functions like decision-making , planning , and impulse control , develops gradually over the first few years of life , with significant maturation occurring during adolescence .

Therefore , the least developed part of the brain at birth is the frontal lobe , which is a key component of the cortex and responsible for higher-order cognitive functions that are not fully functional at birth .

While the cortex is the general term , the frontal lobe is the most specific and relevant answer since it 's the last region to mature .

Among the answer choices , the frontal lobe (C) is explicitly listed and represents the region that develops last at birth .

Hence , the correct answer is C .

< answer > C < / answer >